

$$P(P-I)X = 0$$

PFI

for $p \neq 1$ ~~and~~ $p \neq 0$

$$y = (p - I) x \neq 0$$

for some $x \neq 0$

as $P \neq 0$ $P((P-I)x) = 0$

$$\rho Y = 0$$

$\therefore +$ is ev of P $P \vee =$
 $\therefore 0$ is ev of P

Ref P#6

$$\text{Red } \textcircled{0} \times \neq 0 \quad x \neq 0$$

$$(P - E)(P_X) = 0$$

$$(P - \mathbb{I})v = 0$$

~~$y \neq 0$~~
for some $y \neq 0$

q is ev of p .

$$d = 0, 2, 1$$

$$P = 0 \Rightarrow \text{eig}(P) = \{0\}$$

$$1. P = I \Rightarrow \text{Erg}(P) = \{1\}$$

if $P \neq 0, P \neq I$ then $\text{rank}(P) \neq \infty \Rightarrow \exists v \in V$ s.t. $Pv = 0$
 $\Rightarrow (I - P)v \neq 0$ $\exists P(v - P(v)) = 0$ $v - P(v) \neq 0$
 $(I - P)(v) \neq 0$

Anan malye.

Geometric multiplicity: $g_m(d) = \dim(\text{Nul}(A - dI))$

Algebraic multiplicity: $\text{am}(d) =$ multiplicity of d as a root of the $P_A(d)$

$$d-m(d) \geq \text{grn}(d)$$

proof: $k = g \cdot n(d) = \dim(\text{Null}(A - dI))$
 $\text{Null}(A - dI) \subseteq \mathbb{H}^n$

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Let $\{v_1, \dots, v_k\}$ basis of $\text{Null}(A - dI)$.

Extended basis $\{v_1, \dots, v_k, u_1, \dots, u_{n-k}\}$ be basis of $\mathbb{R}^n$.

Costant matrix $P = [v_1, \dots, v_k, v_1, \dots, v_{n-k}]_{n \times m}$

$$AP = A \begin{bmatrix} - \\ \\ \\ \end{bmatrix} = [AV_1 | AV_k | AV_{k+1} | \dots | AV_n] \cdot$$

$$= \left(\frac{dV_1}{dV_2} \dots \frac{dV_k}{dV_k} \right) A_{V_1} \dots A_{V_k}$$

$$= (P(dq_1) \cdots P(dq_k) | P\tilde{P}'AV_1| \cdots | P\tilde{P}'AV_{m_k})$$

$$P_{el} = V_i$$

$$P(d\epsilon) = dV_i$$

$$AP = P[d_1 \dots d_k | d_{k+1} \dots d_n | P^{-1}A U_1 \dots P^{-1}A U_{n-k}]$$

$$P^{-1}AP = \begin{bmatrix} d_1 & & & * \\ & \ddots & & \\ & & d_k & * \\ 0 & & & \times \end{bmatrix}_{n \times n}$$

$$P_A(\lambda) = \det(\lambda I - A) = \det(\lambda I - P^{-1}AP) =$$

$$= \det \left(\lambda I - \begin{bmatrix} d_1 & & & * \\ & \ddots & & \\ & & d_k & * \\ 0 & & & \times \end{bmatrix}_{n \times n} \right)$$

$$= \det \begin{bmatrix} \lambda - d_1 & & & * \\ & \ddots & & \\ & & \lambda - d_k & * \\ 0 & & & \times \end{bmatrix}$$

$$= (\lambda - d)^k q(\lambda) \quad \text{deg of } q(\lambda) = n - k.$$

$$\therefore a: m(d) \geq k = \text{gm}(d)$$

Exercise: e-vectors corresponding to distinct e-values are L.I.

Let $v_1, \dots, v_k$ be e.v. $\alpha = d_1, \dots, d_n$
 say $v_1, \dots, v_k$ are L.I. $v_{k+1}, \dots, v_n$ are L.D. $\Rightarrow \sum_{i=1}^n c_i v_i = 0$

problem: Let $A_{n \times n}$ s.t. every ^{non-zero} vector is an eigenvector of A .
 then show that $A = \alpha I$, $\alpha \in \mathbb{R}$.

$$\Rightarrow AV_1 = \alpha V_1, AV_2 = \beta V_2$$

take v_1 & v_2 L.I.

$$A(v_1 + v_2) = \gamma(v_1 + v_2)$$

$$AV_1 + AV_2$$

$$\alpha v_1 + \beta v_2 = \gamma v_1 + \gamma v_2$$

$$(\alpha - \gamma)v_1 + (\beta - \gamma)v_2 = 0$$

$$\text{then } \alpha - \gamma \neq 0 \text{ and } \beta - \gamma \neq 0$$

$$\alpha = \gamma \text{ \& } \beta = \gamma$$

$$A e_i = d_i e_i = \alpha e_i$$

$$A = (d_1 e_1 \dots d_n e_n)$$

$$A(e_i + e_j) = d_i(e_i + e_j) = d_i e_i + d_j e_j$$

$$A e_i + A e_j = d_i e_i + d_j e_j$$

$$d_i e_i + d_j e_j = d_i e_i + d_j e_j$$

$$(d_i - d) e_i + (d_j - d) e_j = 0$$

$$e_i \& e_j \text{ are L.I.}$$

$$d_i - \alpha = 0$$

$$d_j - \alpha = 0$$

Diagonalizability matrix:

A $n \times n$ over F

A is diagonalizable iff $\exists P_{n \times n}$ invertible.

s.t. $P^{-1}AP = D$.

Diagonalizable operator:

$T \in L(V)$, T is diagonalizable

iff $\exists$ basis B of V s.t. $[T]_B$ is diagonal.

$$\frac{1}{ad-bc} \begin{pmatrix} d & -c \\ -b & a \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} 0 & d \\ 0 & -b \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\frac{1}{ad-bc} \begin{pmatrix} cd & d^2 \\ -bc & -bd \end{pmatrix} = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}$$

~~but~~

$$d=0, bc=0$$

$$\text{but } ad-bc \neq 0 \quad d=0 \Rightarrow bc \neq 0 \quad \times$$

$$bc=0 \Rightarrow ad \neq 0 \Rightarrow d \neq 0$$

$$\text{if } bc=0 \quad ad$$

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

det. D then $\text{e.v. are } 0, 0$.

$$\exists P \quad P^{-1}AP = D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$P^{-1}AP = 0$$

$$A = 0 \quad \text{contradict.}$$

Result: let A $n \times n$ matrix. then A is diag. iff $\exists$ basis of $\mathbb{R}^n$ consisting the e.vectors of A .

$\Rightarrow$ let A is diag. then by defn $\exists P$ invertible $\in \mathbb{R}^{n \times n}$

$$\text{s.t. } P^{-1}AP = D = \begin{pmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{pmatrix}$$

$$AP = PD$$

$$A[P_1 \ P_2 \ \dots \ P_n] = [P_1 \ P_2 \ \dots \ P_n]D$$

$$[AP_1 \ \dots \ AP_n] = [P_1d_1 \ \dots \ P_nd_n]$$

$$AP_i = d_i P_i$$

d_i are e.v. of A .

$\therefore P_i$ are e.v. forming basis for $\mathbb{R}^n$ (P_i are l.i.)

(look course we go through)